

Homework 2

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Problem 2.1.

Solution 2.1.

1. Intersect at $\begin{bmatrix} 0 \\ 2 \end{bmatrix}$. 2. Intersect at $\begin{bmatrix} 3 \\ 0 \end{bmatrix}$. 3. Parallel.

Problem 2.2. Recall that, whenever we have a degree one equation on two variables, then its solution set is a line on the Cartesian plane $\mathbb{R}^2$. When you draw lines, make sure to find the x, y -intercepts and mark it on the graph (i.e., the intersection of the line with x, y -axes).

For this problem, we want to investigate the following question: what would happen to the lines if we perform linear combinations of equations?

1. Draw the lines for the equations $2x + 3y = 6$, $4x + 6y = 12$, $6x + 9y = 18$. What are their relations?
2. Find the intersection of the lines $2x + y = 4$ and $2x + 3y = 6$.
3. Let us see what happen if we add the two line equations. Draw the lines $2x + y = 4$, $2x + 3y = 6$, and $(2x + y) + (2x + 3y) = 4 + 6$. What are their relations?
4. Draw the lines $(2x + y) + (2x + 3y) = 4 + 6$, $(2x + y) + 2(2x + 3y) = 4 + 12$, $(2x + y) + 3(2x + 3y) = 4 + 18$. What is the relation between them? (Challenge problem: How would you describe the tendency of change as we add more and more $2x + 3y = 6$ to our equation?)

Solution 2.2.

1. They are actually the same line. 2. $\begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$
3. They intersect at the same point $\begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$. And the new line lays between the two lines in the sub-problem 2.
4. Intersect at the same point. The line will approach to $2x + 3y = 6$ as we add more and more $2x + 3y = 6$ to the equation.

Problem 2.3.

Solution 2.3.

1. $\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$. 2. $\begin{bmatrix} 1 & 0 & \frac{2}{9} & -\frac{1}{9} \\ 0 & 1 & \frac{3}{4} & \frac{4}{9} \end{bmatrix}$. 3. $\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$. 4. $\begin{bmatrix} 1 & & \\ & 1 & \\ & & 1 \end{bmatrix}$.
5. $\begin{bmatrix} 1 & & \\ & 1 & \\ & & 1 \end{bmatrix}$. 6. $\begin{bmatrix} 1 & & \\ & 1 & \\ & & 1 \end{bmatrix}$. 7. $\begin{bmatrix} 1 & & \\ & 1 & \\ & & 1 \end{bmatrix}$.

Problem 2.4. Consider the matrix $\begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$. Only use row scaling and row shearing, can you turn it into $\begin{bmatrix} 3 & 5 \\ 1 & 2 \end{bmatrix}$?

Solution 2.4.

Of course we can!

$$\begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix} \xrightarrow{R_1+R_2} \begin{bmatrix} 1 & 2 \\ 4 & 7 \end{bmatrix} \xrightarrow{-R_2+R_1} \begin{bmatrix} -3 & -5 \\ 4 & 7 \end{bmatrix} \xrightarrow{R_1+R_2} \begin{bmatrix} -3 & -5 \\ 1 & 2 \end{bmatrix} \xrightarrow{-1 \times R_1} \begin{bmatrix} 3 & 5 \\ 1 & 2 \end{bmatrix}$$

In fact, row permutation is composition of row scaling and row shearing.

Problem 2.5. (Gaussian Eliminations) Solve these by Gaussian elimination, and write out the solution set explicitly. (So don't just stop at RREF for this problem. Write out the set.)

1. $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \\ 7 \end{bmatrix}$.

$$2. \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \end{bmatrix}.$$

$$3. \begin{bmatrix} 1 & 2 \\ 4 & 5 \\ 7 & 8 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \\ 8 \end{bmatrix}.$$

Solution 2.5.

$$1. \text{ The RREF of the auxiliary matrix is } \left[\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]. \text{ Solution set: } \left\{ \begin{bmatrix} 1+z \\ -2z \\ z \end{bmatrix}, z \in \mathbb{R} \right\} \text{ or } \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + k \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}, k \in \mathbb{R}.$$

2. Same as 1.

3. Solution set $\emptyset$.

Problem 2.6.

Solution 2.6.

1.
The equation has no solution $\iff$ The matrix is not surjective $\iff$ rows of the matrix are linearly dependent $\iff b = 6$.

Another approach is to use REF, $\left[\begin{array}{cc|c} 1 & 2 & 3 \\ 3 & b & 7 \end{array} \right] \rightarrow \left[\begin{array}{cc|c} 1 & 2 & 3 \\ 0 & b-6 & -2 \end{array} \right]$. No solution if and only if $b = 6$.

2.
Since the two row vectors have been parallel to each other, the equation has no solution if and only if they are not overlap, that is, $b \neq 20$.

Another approach is to use REF, $\left[\begin{array}{cc|c} 3 & 2 & 10 \\ 6 & 4 & b \end{array} \right] \rightarrow \left[\begin{array}{cc|c} 3 & 2 & 10 \\ 0 & 0 & b-20 \end{array} \right]$. No solution if and only if $b \neq 20$.

3.
The equation has no solution $\iff$ The matrix is not surjective $\iff$ rows of the matrix are linearly dependent $\iff b = 11$.

Another approach is to use REF, $\left[\begin{array}{ccc|c} 2 & 5 & 1 & 0 \\ 4 & b & 1 & 2 \\ 0 & 1 & -1 & 3 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 2 & 5 & 1 & 0 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & b-11 & 32-3b \end{array} \right]$. No solution if and only if $b = 11$.

4.
The equation has infinite solution $\iff$ The matrix is not injective $\iff$ columns of the matrix are linearly dependent $\iff b = -3$.

Another approach is to use REF, $\left[\begin{array}{cc|c} b & 3 & 6 \\ 3 & b & -6 \end{array} \right] \rightarrow \left[\begin{array}{cc|c} 3 & b & -6 \\ 0 & 3 - \frac{b^2}{3} & 6 + 2b \end{array} \right]$. Hence it has infinite solution if and only if $3 - \frac{b^2}{3} = 6 + 2b = 0 \implies b = -3$.

5.
The equation has infinite solution $\iff$ The matrix is not injective $\iff$ columns of the matrix are linearly dependent $\iff b = 4$. In this case, we must have $c = 32$ otherwise there will be no solution.

Another approach is to use REF, $\left[\begin{array}{cc|c} 2 & b & 16 \\ 4 & 8 & c \end{array} \right] \rightarrow \left[\begin{array}{cc|c} 2 & b & 16 \\ 0 & 8-2b & c-32 \end{array} \right]$. Hence it has infinite solution if and only if $8-2b = c-32 = 0 \implies b = 4, c = 32$.

6.
The equation has infinite solution $\iff$ The matrix is not injective $\iff$ columns of the matrix are linearly dependent $\iff b = -1$.

Another approach is to use REF, $\left[\begin{array}{ccc} 1 & b & 0 \\ 1 & -2 & -1 \\ 0 & 1 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc} 1 & b & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1+b \end{array} \right]$. Hence it has non-zeros solutions if and only if $b = -1$.

7. Using REF, $\begin{bmatrix} b & 2 & 3 \\ b & b & 4 \\ b & b & b \end{bmatrix} \rightarrow \begin{bmatrix} b & 2 & 3 \\ 0 & b-2 & 1 \\ 0 & 0 & b-4 \end{bmatrix}$. Hence it has non-zero solutions if and only if $b = 0, 2, 4$.

Problem 2.7. For each real value of p , find all solutions to the following system:

$$\begin{bmatrix} p & 1 & 1 \\ 1 & p & 1 \\ 1 & 1 & p \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ p \\ p^2 \end{bmatrix}.$$

(Hint: first replace the first row by the sum of three rows.)

Solution 2.7.

$$\left[\begin{array}{ccc|c} p & 1 & 1 & 1 \\ 1 & p & 1 & p \\ 1 & 1 & p & p^2 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} p+2 & p+2 & p+2 & p^2+p+1 \\ 1 & p & 1 & p \\ 1 & 1 & p & p^2 \end{array} \right] \quad (1)$$

Assumed that $p+2 \neq 0$, we have

$$\left[\begin{array}{ccc|c} p+2 & p+2 & p+2 & p^2+p+1 \\ 1 & p & 1 & p \\ 1 & 1 & p & p^2 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 1 & \frac{p^2+p+1}{p+2} \\ 1 & p & 1 & p \\ 1 & 1 & p & p^2 \end{array} \right] \quad (2)$$

$$\rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 1 & \frac{p^2+p+1}{p+2} \\ 0 & p-1 & 0 & \frac{p-1}{p+2} \\ 0 & 0 & p-1 & \frac{p^3+p^2-p-1}{p+2} \end{array} \right] \quad (3)$$

Note that $p^3+p^2-p-1 = (p+1)^2(p-1)$, and we also assume $p-1 \neq 0$.

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & \frac{p^2+p+1}{p+2} \\ 0 & p-1 & 0 & \frac{p-1}{p+2} \\ 0 & 0 & p-1 & \frac{p^3+p^2-p-1}{p+2} \end{array} \right] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & \frac{p^2+p+1}{p+2} \\ 0 & p-1 & 0 & \frac{p-1}{p+2} \\ 0 & 0 & p-1 & \frac{(p+1)^2(p-1)}{p+2} \end{array} \right] \quad (4)$$

$$\rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 1 & \frac{p^2+p+1}{p+2} \\ 0 & 1 & 0 & \frac{1}{p+2} \\ 0 & 0 & 1 & \frac{(p+1)^2}{p+2} \end{array} \right] \quad (5)$$

$$\rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & \frac{-(p+1)}{p+2} \\ 0 & 1 & 0 & \frac{1}{p+2} \\ 0 & 0 & 1 & \frac{(p+1)^2}{p+2} \end{array} \right] \quad (6)$$

Therefore,

(a) if $p \neq -2$ and $p \neq 1$, from (6), the equation has a unique solution $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \frac{-(p+1)}{p+2} \\ \frac{1}{p+2} \\ \frac{(p+1)^2}{p+2} \end{bmatrix}.$

(b) if $p = -2$, from (1), the solution set is $\emptyset$.

(c) if $p = 1$, from (4), the solution set is $\left\{ \begin{bmatrix} 1 - y - z \\ y \\ z \end{bmatrix}, y, z \in \mathbb{R} \right\}$, or $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + y \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + z \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, y, z \in \mathbb{R}$.

Problem 2.8. (Rank-one matrices) If a matrix has all rows parallel, must it also have all columns parallel? (As a convention, we treat $\mathbf{0}$ as parallel to everything.) Give a counter-example or prove why.

Solution 2.8.

If a matrix has all rows parallel, all of its columns are also parallel.

Proof:

Let's say an arbitrary $m \times n$ matrix A has all rows parallel to vector $\mathbf{a}$. Set $\mathbf{a} = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} \in \mathbb{R}^n$. Hence

$$A = \begin{bmatrix} k_1 \mathbf{a}^T \\ k_2 \mathbf{a}^T \\ \vdots \\ k_m \mathbf{a}^T \end{bmatrix} = \begin{bmatrix} k_1 \\ k_2 \\ \vdots \\ k_m \end{bmatrix} \mathbf{a}^T$$

Set $\mathbf{b} = \begin{bmatrix} k_1 \\ k_2 \\ \vdots \\ k_m \end{bmatrix}$, we have

$$A = \begin{bmatrix} k_1 \\ k_2 \\ \vdots \\ k_m \end{bmatrix} \mathbf{a}^T = \mathbf{b} \mathbf{a}^T = \mathbf{b} \begin{bmatrix} a_1 & a_2 & \cdots & a_n \end{bmatrix} = \begin{bmatrix} a_1 \mathbf{b} & a_2 \mathbf{b} & \cdots & a_n \mathbf{b} \end{bmatrix}$$

So all of its columns are also parallel.